

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Artificial Intelligence and Data Science

Paper ID

1320

Paper Title

Weighted Ensemble Steganalysis: Integrating CNN, XGBoost, and LightGBM

Abstract

Digital image steganography poses a significant challenge to digital forensics due to the microscopic nature of hidden data. Existing detection methods typically rely either on handcrafted statistical features (Traditional Steganalysis) or end-to-end deep learning models (Steganalysis CNNs). This paper proposes a novel Weighted Heterogeneous Ensemble Model (WHEM) that synergistically integrates both approaches to maximize detection accuracy and robustness. The WHEM combines the prediction probabilities from:

- 1) A specialized Steganalysis Convolutional Neural Network (CNN) structure, derived from a GAN Discriminator, operating on image residual noise;
- 2) XGBoost, and
- 3) LightGBM, both operating on statistical domain features.

Crucially, the final prediction is determined by a weight optimization search that maximizes the overall F1-score. Experimental results on a simulated dataset demonstrate that this heterogeneous ensemble achieves a classification accuracy of $\approx 74.50\%$, a statistically significant improvement over all individual component models. This proves that combining models with complementary views of the data—raw noise vs. statistical patterns—is an effective strategy for robust steganalysis.

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Submission Questions Response

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